

OSPF (Part-3)

- Topics covered:
 - OSPF Network Type
 - OSPF Neighbour/Adjacency Requirements
 - OSPF LSA Types

Loopback Interface

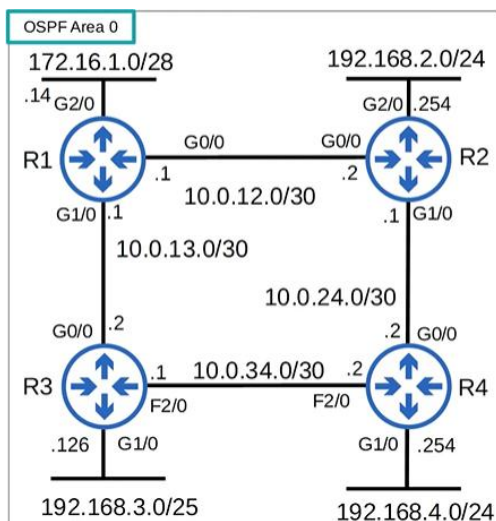
A loopback interface is a virtual interface on a router.

- It is always up/up, unless manually shut down.
- Its status does not depend on physical interfaces.
- Physical interfaces can fail due to hardware issues.
- A loopback interface will only fail if the router itself fails.
- It provides a consistent IP address to:
 - Identify the router
 - Reach the router

Sometimes traffic must be sent directly to a router, not just through it.

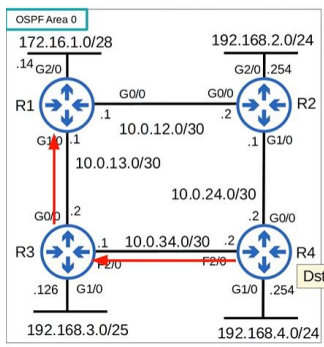
Scenario: No Loopback Interface on R1:

Destination: 10.0.13.1 (R1 G1/0)

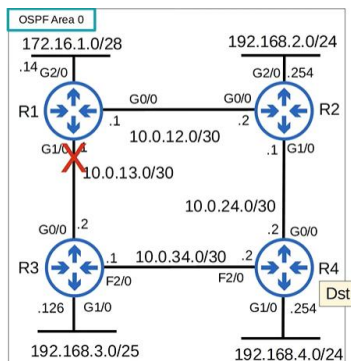


Dst: 10.0.13.1

- R4 forwards traffic toward R1's G1/0 interface (10.0.13.1).



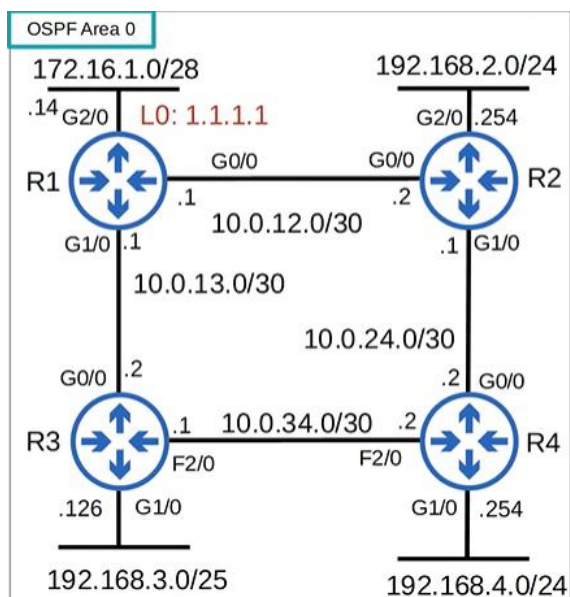
- If R1 G1/0 goes down, traffic to 10.0.13.1 fails.



- R4 can no longer reach R1 using that address.

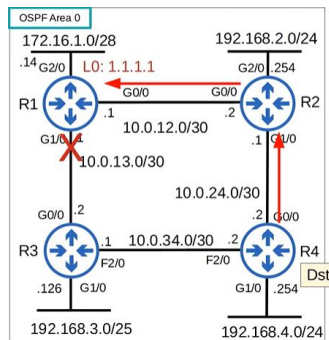
Scenario: Loopback Interface on R1:

Loopback: 1.1.1.1



Dst: 1.1.1.1

- Even if a physical interface fails:
 - Traffic to **1.1.1.1** can still reach R1.



- Loopback provides:
 - A stable identifier
 - A consistently reachable IP address

Conclusion:

Configuring a loopback interface is a good practice because it provides an IP address that is always up and can consistently identify and reach the router.

OSPF Network Types

How Many Types of OSPF Network?

The OSPF network type refers to the type of connection between OSPF neighbours. This type influences OSPF behaviour.

Three Main OSPF Network Types

1. Broadcast
2. Point-to-Point
3. Non-Broadcast

Broadcast Network Type:

- Enabled by default on:
 - Ethernet
 - FDDI
- FDDI is an old technology.
- You only need to remember:
 - OSPF uses the Broadcast network type by default on it.

Point-to-Point Network Type:

- Enabled by default on:
 - PPP
 - HDLC
- You do not need deep knowledge of these for CCNA.

Non-Broadcast Network Type

- Enabled by default on:
 - Frame Relay
 - X.25
- Neighbours must be manually configured.
- Included for exam awareness only.

CCNA Exam Focus

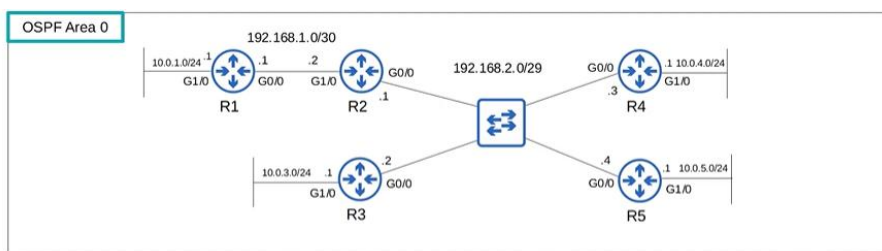
- **3.4b** – Point-to-point network type
- **3.4c** – Broadcast network type

3.4	Configure and verify single area OSPFv2
3.4.a	Neighbor adjacencies
3.4.b	Point-to-point
3.4.c	Broadcast (DR/BDR selection)
3.4.d	Router ID

These are the network types focused on in this section.

Broadcast Network Type – Ethernet

All connections shown use Gigabit Ethernet (G).



Because all links are Ethernet:

- All use the Broadcast network type.

Characteristics of the Broadcast Network Type:

Neighbour Discovery-

- Routers dynamically discover neighbours using OSPF Hello messages.
- Multicast address used:
 - 224.0.0.5
- Not all network types do this.

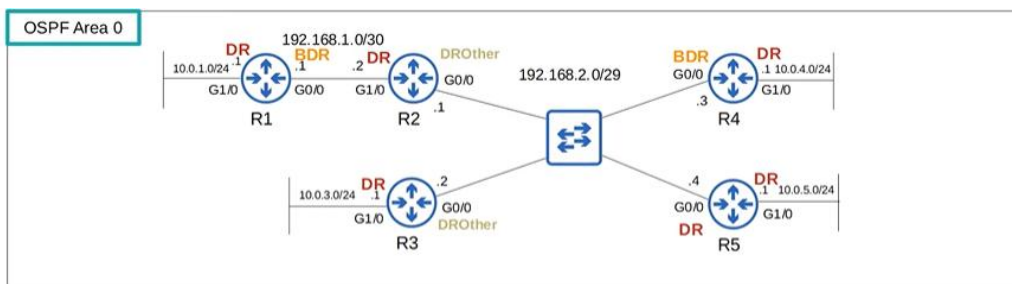
DR and BDR Election-

- A DR and BDR must be elected on each subnet.
- Exception:
 - If there are no OSPF neighbours, only a DR exists.

Routers that are neither DR nor BDR become:

- DROther

Example Network with DR/BDR:



- Each subnet needs a DR.
- Subnets with only one router:
 - That router becomes the DR.
- Multi-router subnet:
 - One DR
 - One BDR
 - Others are DROthers

How DR and BDR Are Elected:

Election Order

1. Highest OSPF interface priority
 2. If equal:
 - Highest OSPF router ID
- Default interface priority: 1
 - Highest router ID wins if priorities match.

SHOW IP OSPF INTERFACE (DR Example):

```
R5#show ip ospf interface g0/0
GigabitEthernet0/0 is up, line protocol is up
Internet Address 192.168.2.4/29, Area 0, Attached via Network Statement
Process ID 1, Router ID 5.5.5.5, Network Type BROADCAST, Cost: 1
Topology-MTID      Cost      Disabled      Shutdown      Topology Name
  0                1         no           no           Base
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 5.5.5.5, Interface address 192.168.2.4
Backup Designated router (ID) 4.4.4.4, Interface address 192.168.2.3
```

- Router ID comes from the loopback interface.
- R5 has the highest router ID.
- R5 becomes the DR.

DR Other Example:

```
R2#show ip ospf int g0/0
GigabitEthernet0/0 is up, line protocol is up
Internet Address 192.168.2.1/29, Area 0, Attached via Network Statement
Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
Topology-MTID      Cost      Disabled      Shutdown      Topology Name
  0                1         no           no           Base
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 5.5.5.5, Interface address 192.168.2.4
Backup Designated router (ID) 4.4.4.4, Interface address 192.168.2.3
```

- R2 is neither DR nor BDR.
- Therefore, it is DROTHER.

Changing OSPF Interface Priority:

```
R2(config)#int g0/0
R2(config-if)#ip ospf priority ?
<0-255> Priority
R2(config-if)#ip ospf priority 255
```

- Range: 0 to 255
- Priority 0:
 - Router cannot become DR or BDR.

After Changing Priority (Non-Pre-emptive Behaviour):

State DROTHER, Priority 255

- Even with highest priority:
 - Role does not change.

```

R2#show ip ospf int g0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 192.168.2.1/29, Area 0, Attached via Network Statement
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
    0                1          no            no            Base
  Transmit Delay is 1 sec, State DROTHER, Priority 255
  Designated Router (ID) 5.5.5.5, Interface address 192.168.2.4
  Backup Designated router (ID) 4.4.4.4, Interface address 192.168.2.3

```

Reason:

DR/BDR election is non-pre-emptive.

Resetting the OSPF Process:

```

R5#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
R5#
*Aug 22 04:25:05.307: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Aug 22 04:25:05.311: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Aug 22 04:25:05.311: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

```

Neighbour Down: Interface down or detached

Then R2 and R4 return state FULL to DOWN

```

*Aug 22 04:25:13.903: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
*Aug 22 04:25:13.907: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

```

- Neighbours go DOWN.
- Adjacencies reform.

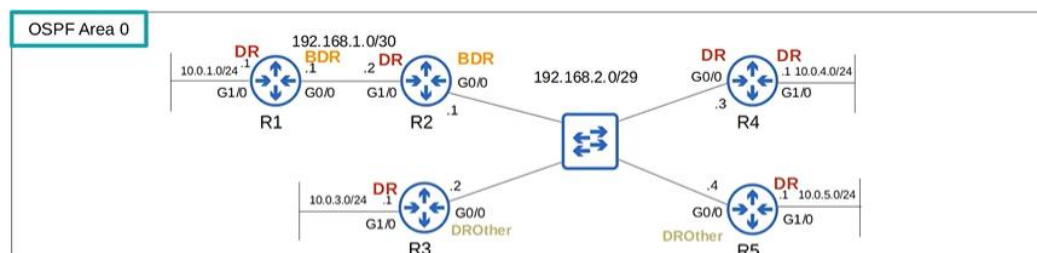
SHOW IP OSPF NEIGHBOR:

```

R5#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	255	FULL/BDR	00:00:37	192.168.2.1	GigabitEthernet0/0
3.3.3.3	1	2WAY/DROTHER	00:00:37	192.168.2.2	GigabitEthernet0/0
4.4.4.4	1	FULL/DR	00:00:39	192.168.2.3	GigabitEthernet0/0



Key Observations

1. BDR becomes DR when DR goes down.
2. A new BDR is elected.
3. DROTHERs:
 - Stay in 2-Way state with other DROTHERs.
 - Reach FULL state only with DR and BDR.

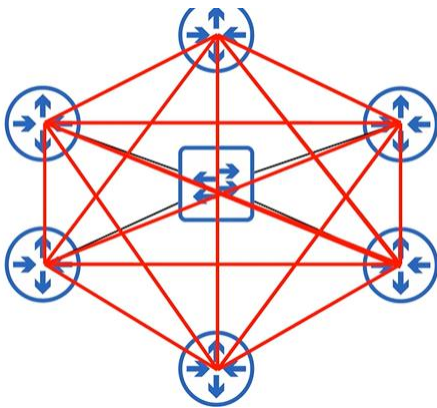
Purpose of DR and BDR:

- DROthers form full adjacencies only with DR and BDR.
- LSAs are exchanged:
 - Only with DR and BDR
- DROthers do not exchange LSAs with each other.
- All routers still end up with the same LSDB.

LSA Flooding Without DR/BDR:

All routers exchanging LSAs with all routers

(Dense full mesh of flooding)

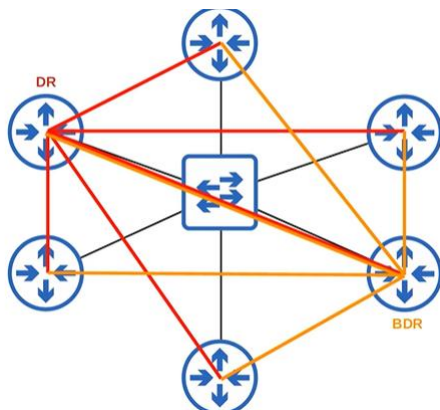


LSA Flooding With DR/BDR:

Routers -> DR / BDR -> Routers

(Reduced flooding)

- Fewer LSAs
- Less congestion
- Reduced unnecessary traffic



Multicast Addresses Used:

- **224.0.0.5** → All OSPF routers
- **224.0.0.6** → DR and BDR only

OSPF Neighbour Process (Review):

The first steps of the OSPF process involve becoming neighbors.

When routers reach this point, they are considered OSPF neighbors.

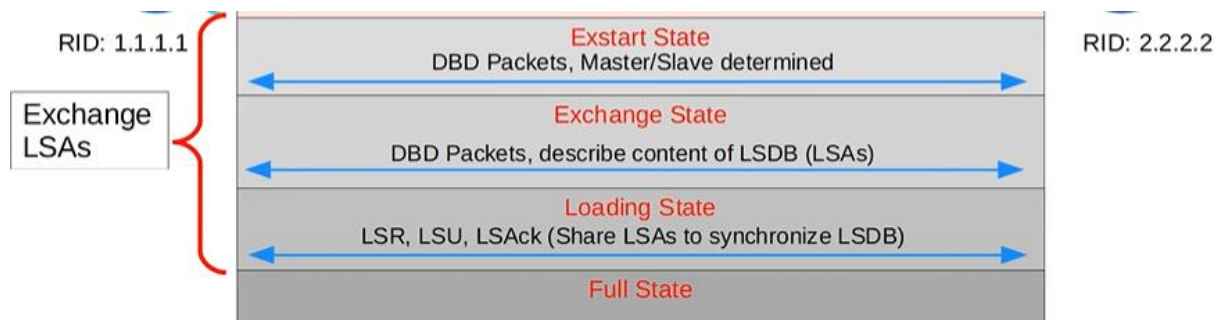
Connections between two DROthers stop at this point.



Routers will only continue to:

- Exchange LSAs
- Form a Full OSPF adjacency

with the DR and BDR only.



Summary of Adjacency Rules:

- The DR and BDR form a Full adjacency with ALL routers in the subnet, including DROthers.
- DROthers form a Full adjacency only with the DR and BDR.
- DROthers do not form Full adjacencies with other DROthers.

SHOW IP OSPF INTERFACE BRIEF (DROther Example):

Command shown on R3, which is a DROther.

```
R3#show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Gi0/0	1	0	192.168.2.2/29	1	DROTH	2/3
Gi1/0	1	0	10.0.3.1/24	1	DR	0/0

Explanation

- F indicates the number of Full adjacencies.
- C indicates the total number of neighbors.

On G0/0:

- Full adjacencies = 2
- Total neighbors = 3

So R3 has:

- Full adjacencies with R2 and R4
- Total neighbors R2, R4, and R5

SHOW IP OSPF INTERFACE G0/0 (Detailed View):

```
R3#show ip ospf interface g0/0
GigabitEthernet0/0 is up, line protocol is up
Internet Address 192.168.2.2/29, Area 0, Attached via Network Statement
Process ID 1, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 1
Topology-MTID Cost Disabled Shutdown Topology Name
0 1 no no Base
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 4.4.4.4, Interface address 192.168.2.3
Backup Designated router (ID) 2.2.2.2, Interface address 192.168.2.1
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
oob-resync timeout 40
Hello due in 00:00:09
Supports Link-local Signaling (LLS)
Cisco NSF helper support enabled
IETF NSF helper support enabled
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 1
Last flood scan time is 0 msec, maximum is 4 msec
Neighbor Count is 3, Adjacent neighbor count is 2
  Adjacent with neighbor 2.2.2.2 (Backup Designated Router)
  Adjacent with neighbor 4.4.4.4 (Designated Router)
Suppress hello for 0 neighbor(s)
```

Explanation

- Neighbour Count is 3 → total OSPF neighbours.
- Adjacent neighbour count is 2 → Full adjacencies.
- Full adjacencies are with:
 - R2 (BDR)
 - R4 (DR)

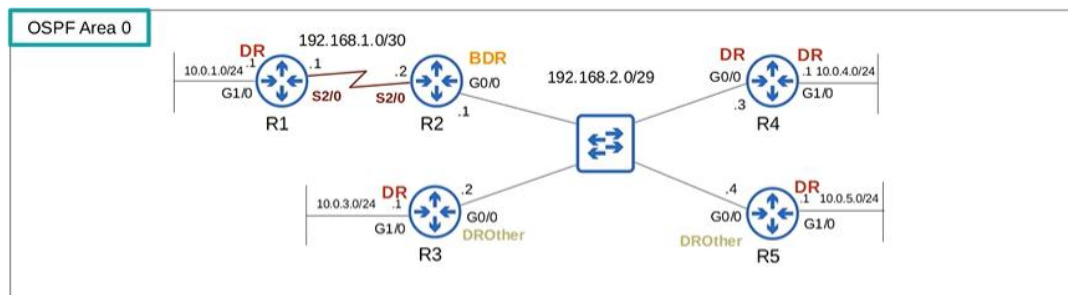
Transition to Point-to-Point Network Type:

That concludes the Broadcast network type section.

Now we move on to the Point-to-Point network type.

Point-to-Point Network Type

The connection between R1 and R2 has been changed to a serial connection.



Point-to-Point Basics:

- Enabled by default on serial interfaces using:
 - PPP
 - HDLC
- PPP and HDLC are Layer 2 encapsulations.
- Similar to Ethernet, but used on serial connections.

Neighbour Discovery:

- Routers dynamically discover neighbours.
- OSPF Hello messages use:
 - 224.0.0.5

Same as Broadcast network type.

Key Difference from Broadcast:

- NO DR and NO BDR election

Reason:

- These encapsulations are used for point-to-point connections.
- There are only two routers.
- There is no point in electing a DR or BDR.

Result:

- The two routers form a Full adjacency directly.

Very Brief Overview of Serial Connections:

Serial connections are:

- Old technology
- Less common today
- Still exist
- Ethernet is much more dominant

Serial interfaces are no longer tested directly, except for:

- OSPF point-to-point network type

Basic understanding is required.

Serial Interfaces and Cables:



- Ports and cables are different from Ethernet.

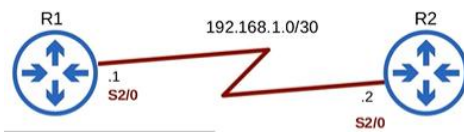
DCE and DTE Roles:

- One side functions as **DCE** (Data Communications Equipment).
- The other side functions as **DTE** (Data Terminal Equipment).

Why this matters:

- The DCE side must specify the clock rate.
- Clock rate = speed of the connection.

Configuring R1 Serial Interface (DCE Side):



```
R1(config)#interface s2/0
R1(config-if)#clock rate ?
    With the exception of the following standard values not subject to rounding,

    1200 2400 4800 9600 14400 19200 28800 38400
    56000 64000 128000 2015232

    accepted clockrates will be bestfitted (rounded) to the nearest value
    supportable by the hardware.

    <246-8064000>    DCE clock rate (bits per second)
```

```
R1(config-if)#clock rate 64000
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
```

- Clock rate configured: 64000 bits per second
- IP address configured
- Interface enabled

Important distinction:

- Ethernet uses SPEED
- Serial uses CLOCK RATE

SHOW INTERFACE S2/0 (Default Encapsulation):

```
R1#show interface s2/0
Serial2/0 is up, line protocol is up
Hardware is M4T
Internet address is 192.168.1.1/24
MTU 1500 bytes, BW 1544 Kbit/sec, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC crc 16, loopback not set
```

- Default encapsulation is HDLC
- On Cisco routers, this is ChdLC

cHDLC frame structure [edit]

The following table describes the structure of a cHDLC frame on the wire. [citation needed]

Address	Control	Protocol Code	Information	Frame Check Sequence (FCS)	Flag
8 bits	8 bits	16 bits	Variable length, 0 or more bits, in multiples of 8	16 bits	8 bits

- The Address field is used to specify the type of packet contained in the cHDLC frame; 0x0F for Unicast and 0x8F for Broadcast packets.
- The Control field is always set to zero (0x00).
- The Protocol Code field is used to specify the protocol type encapsulated within the cHDLC frame (e.g. 0x0800 for Internet Protocol).

- Displayed as HDLC in CLI
- It is a Layer 2 encapsulation

PPP Encapsulation Configuration:

```
R1(config)#int s2/0
R1(config-if)#encapsulation ppp
```

Important note:

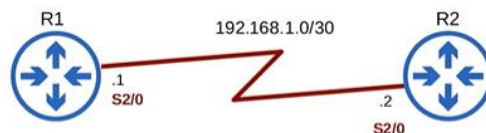
- Encapsulation must match on both ends
- If different:
 - Interface will go down
 - Routers cannot communicate

SHOW INTERFACE After PPP:

```
R1(config-if)#do show interface s2/0
Serial2/0 is up, line protocol is up
Hardware is M4T
Internet address is 192.168.1.1/24
MTU 1500 bytes, BW 1544 Kbit/sec, DLY 20000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation PPP, LCP Open
```

- Encapsulation successfully changed to PPP

Running Configuration (R1 and R2):



```
R1#show running-config interface s2/0
Building configuration...

Current configuration : 126 bytes
!
interface Serial2/0
ip address 192.168.1.1 255.255.255.0
encapsulation ppp
serial restart-delay 0
clock rate 64000
end
```

```
R2#show running-config interface s2/0
Building configuration...

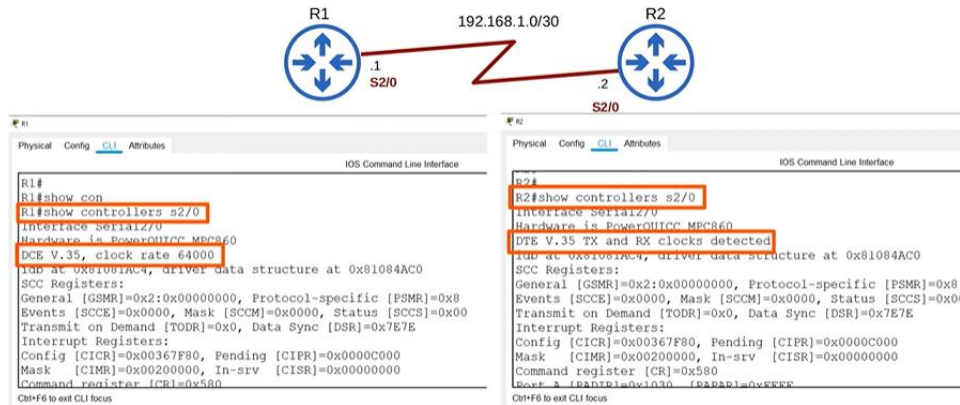
Current configuration : 110 bytes
!
interface Serial2/0
ip address 192.168.1.2 255.255.255.252
encapsulation ppp
serial restart-delay 0
end
```

- R2 has no clock rate
- R2 is the DTE side

Identifying DCE vs DTE:

Command used:

show controllers <interface>



- R1 is the DCE
- Clock rate is configured locally
- R2 is DTE
- Receives clocking from R1

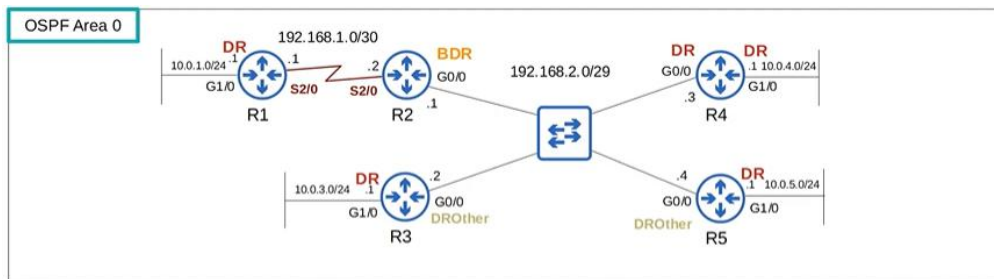
Serial Connection Summary:

- Default serial encapsulation is HDLC
- PPP can be configured with:
- encapsulation ppp
- Encapsulation must match on both ends
- One side is DCE, the other is DTE
- Use:
- show controllers <interface>

to identify DCE/DTE

- Configure CLOCK RATE on the DCE side only
- Clock rate is specified in bits per second

SHOW IP OSPF NEIGHBOR on R2 (Point-to-Point):



```
R2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	0	FULL/-	00:00:31	192.168.1.1	Serial2/0
3.3.3.3	1	2WAY/DROther	00:00:39	192.168.2.2	GigabitEthernet0/0
4.4.4.4	1	FULL/DR	00:00:38	192.168.2.3	GigabitEthernet0/0
5.5.5.5	1	FULL/BDR	00:00:31	192.168.2.4	GigabitEthernet0/0

Explanation

- R2 has a Full adjacency with R1 on the serial link.
- Instead of DR, BDR, or DROther, a dash (-) is displayed.
- This is because the point-to-point network type does not use DRs or BDRs.

Manually Configuring the OSPF Network Type:

You can manually configure the OSPF network type on an interface.

Command Syntax: ip ospf network ?

```
R1(config-if)#ip ospf network ?
```

broadcast	Specify OSPF broadcast multi-access network
non-broadcast	Specify OSPF NBMA network
point-to-multipoint	Specify OSPF point-to-multipoint network
point-to-point	Specify OSPF point-to-point network

Why Change the OSPF Network Type?

Example scenario:

R1 ----- R2

G0/0 G0/0

Ethernet

- Two routers are directly connected with an Ethernet link.
- There is no need for a DR or BDR.
- You can configure the interface as point-to-point.
- This is optional, not required.

Network Type Compatibility Note:

- Not all OSPF network types work on all link types.
- Example:
 - A serial link cannot use the broadcast network type.
- Reason:
 - Serial links do not support Layer 2 broadcast frames.
 - Broadcast network type requires Layer 2 broadcast capability.

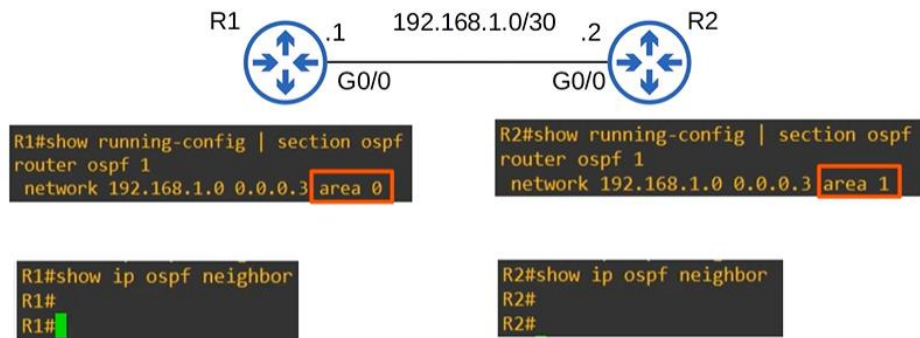
OSPF Network Type Quick Review Chart:

Broadcast	Point-to-point
Default on Ethernet , FDDI interfaces	Default on HDLC , PPP (serial) interfaces
DR/DBR elected	No DR/BDR
Neighbors dynamically discovered	Neighbors dynamically discovered
Default timers: Hello 10, Dead 40	Default timers: Hello 10, Dead 40

(**Non-broadcast** network type default timers = Hello 30, Dead 120)

OSPF Neighbour/Adjacency Requirements

Requirement 1: Area Number Must Match



- No OSPF neighbour's form

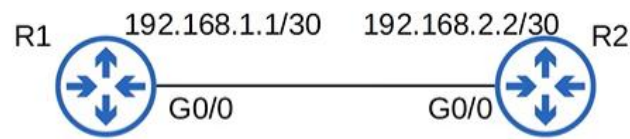
Fix: By Changing (area 1) to same as area of R1 (area 0)



Rule:

- Routers must be in the same area to become OSPF neighbours.

Requirement 2: Interfaces Must Be in the Same Subnet



```
R1#show running-config | section ospf
router ospf 1
 network 192.168.1.0 0.0.0.3 area 0
R1#
```

```
R2#show running-config | section ospf
router ospf 1
 network 192.168.2.0 0.0.0.3 area 0
R2#
```

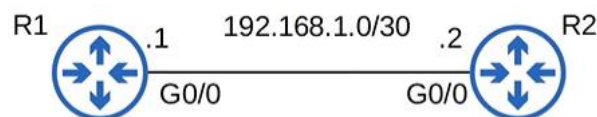
```
R1#show ip ospf neighbor
R1#
R1#
```

```
R2#show ip ospf neighbor
R2#
R2#
```

- OSPF enabled on both interfaces.
- Interfaces are in different subnets.
- No OSPF neighbour's form

Fix:

- R2 interface reconfigured into the same subnet as R1.
- Network command updated accordingly.



```
R1#show running-config | section ospf
router ospf 1
 network 192.168.1.0 0.0.0.3 area 0
R1#
```

```
R2#show running-config | section ospf
router ospf 1
 network 192.168.1.0 0.0.0.3 area 0
R2#
```

```
R1#show ip ospf neighbor
Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.1.2    1     FULL/BDR        00:00:34   192.168.1.2   GigabitEthernet0/0
R1#
```

```
R2#show ip ospf neighbor
Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.1.1    1     FULL/DR         00:00:39   192.168.1.1   GigabitEthernet0/0
R2#
```

Result:

- R1 and R2 become OSPF neighbours again.

Requirement 3: OSPF Process Must Not Be Shutdown

The OSPF process itself can be shut down.

Shutdown OSPF Process on R2:

```
R2(config)#router ospf 1
R2(config-router)#shutdown
R2(config-router)#
*Aug 23 03:43:31.719: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Int
R2(config-router)#do show ip ospf neighbor
R2(config-router)#
```

Result

- Message indicates neighbour went from FULL to DOWN.
- SHOW IP OSPF NEIGHBOR shows no neighbours.

Enable OSPF Again:

```
R2(config-router)#no shutdown
R2(config-router)#
*Aug 23 03:49:52.931: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
R2(config-router)#do show ip ospf neighbor

Neighbor ID    Pri   State       Dead Time   Address        Interface
192.168.1.1    1     FULL/DR     00:00:38   192.168.1.1   GigabitEthernet0/0
R2(config-router)#
```

Result

- Message indicates neighbour is back up.
- Neighbour appears again in SHOW IP OSPF NEIGHBOR.

Requirement 4: Router IDs Must Be Unique

Initial State:

- No loopback interfaces configured.
- Router IDs chosen from interface IPs:
 - R1 → 192.168.1.1
 - R2 → 192.168.1.2

Duplicate Router ID Configuration:

On R2:

```
R2(config-router)#router-id 192.168.1.1
% OSPF: Reload or use "clear ip ospf process" command, for this to take effect
R2(config-router)#end
R2#clear ip
*Aug 23 03:57:58.835: %SYS-5-CONFIG_I: Configured from console by console
R2#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
R2#
*Aug 23 03:58:04.055: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or d
R2#
*Aug 23 03:58:06.495: %OSPF-4-DUP_RTRID_NBR: OSPF detected duplicate router-id 192.168.1.1 from 192.168.1.1 on interface GigabitEthernet0/0
R2#show ip ospf neighbor
R2#
```

Result:

- Neighbour goes DOWN.
- Message displayed:

OSPF detected duplicate router-id 192.168.1.1

from 192.168.1.1 on interface GigabitEthernet0/0

- Neighbour remains down.

Fix Duplicate Router ID:

```
R2(config-router)#no router-id
R2(config-router)#
*Aug 23 04:10:10.207: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
R2(config-router)#do show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.1.1    1     FULL/DR         00:00:35   192.168.1.1   GigabitEthernet0/0
R2(config-router)#
```

- Router ID automatically returns to 192.168.1.2.
- Neighbour comes back without resetting OSPF again.

Reason:

- R2 had no other OSPF neighbours at the time.
- Router could safely change its router ID.

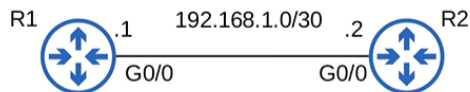
Requirement 5: Hello and Dead Timers Must Match

Defaults (Broadcast & Point-to-Point):

- Hello: 10 seconds
- Dead: 40 seconds

Timer Mismatch Configuration:

On R2 interface:



```
R2(config-if)#ip ospf hello-interval ?
<1-65535> Seconds
R2(config-if)#ip ospf hello-interval 5
R2(config-if)#ip ospf dead-interval ?
<1-65535> Seconds
minimal Set to 1 second
R2(config-if)#ip ospf dead-interval 20
R2(config-if)#
*Aug 23 04:29:30.623: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down:
R2(config-if)#do show ip ospf neighbor
R2(config-if)#
```

Result:

- Neighbour goes DOWN.
- Dead timer expires.

Fix Timers:

```
R2(config-if)#no ip ospf hello-interval
R2(config-if)#no ip ospf dead-interval
R2(config-if)#
*Aug 23 04:31:32.727: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
R2(config-if)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.1	1	FULL/BDR	00:00:35	192.168.1.1	GigabitEthernet0/0

```
R2(config-if)#
```

- Timers return to default values.
- Neighbour comes back UP.

Note

- Changing either Hello or Dead timer alone is enough to break adjacency.
- Use NO commands to return to defaults.
- Exact values do not need to be specified when removing.

Requirement 6: Authentication Settings Must Match

OSPF authentication can be configured so that routers only become neighbours if they have a matching OSPF password.

The OSPF password is configured directly on the interface.

Configuring an OSPF Authentication Key:

CLI Representation (R2):

```
R2(config-if)#ip ospf authentication-key jeremy
R2(config-if)#ip ospf authentication
R2(config-if)#
*Aug 23 04:56:28.435: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: L
R2(config-if)#do show ip ospf neighbor
R2(config-if)#
```

- ip ospf authentication-key configures the password.
- This command does NOT enable authentication by itself.
- ip ospf authentication is required to enable authentication on the interface.

Result: Neighbour Goes Down:

- R1 is not providing a matching OSPF password.
- In fact, authentication was not configured on R1 at all.

Fixing the Authentication Mismatch:

Authentication was removed from R2.

```
R2(config-if)#no ip ospf authentication
R2(config-if)#no ip ospf authentication-key jeremy
*Aug 23 04:59:37.315: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
R2(config-if)#do show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.1.1    1     FULL/BDR        00:00:34    192.168.1.1    GigabitEthernet0/0
R2(config-if)#
```

Result: Neighbour Comes Back Up

Requirement 7: IP MTU Settings Must Match

The IP MTU is the maximum size of an IP packet that can be sent out an interface.

- Default MTU is usually 1500 bytes.
- MTU can be manually configured.

Special Note (Important):

- This requirement (and the next one) is special.
- Even if settings do not match:
 - Routers can become OSPF neighbours
 - But OSPF will not function properly

Changing MTU on R2:

```
R2(config-if)#ip mtu ?  
<68-1500> MTU (bytes)  
R2(config-if)#ip mtu 1400
```

- R2 MTU = 1400
- R1 MTU = 1500

Initial Observation: After waiting

- R1 and R2 remain neighbours

```
R2(config-if)#do show ip ospf neighbor  


| Neighbor ID | Pri | State    | Dead Time | Address     | Interface          |
|-------------|-----|----------|-----------|-------------|--------------------|
| 192.168.1.1 | 1   | FULL/BDR | 00:00:34  | 192.168.1.1 | GigabitEthernet0/0 |


```

Resetting the OSPF Process:

```
R2(config-if)#do clear ip ospf process  
Reset ALL OSPF processes? [no]: yes  
R2(config-if)#  
*Aug 23 05:16:07.474: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
```

Result After Reset:

```
R2(config-if)#do show ip ospf neighbor  


| Neighbor ID | Pri | State      | Dead Time | Address     | Interface          |
|-------------|-----|------------|-----------|-------------|--------------------|
| 192.168.1.1 | 1   | EXSTART/DR | 00:00:38  | 192.168.1.1 | GigabitEthernet0/0 |


```

- No message appears indicating the FULL state.
- Neighbour is stuck in EXSTART.

Repeated Error Messages:

```
*Aug 23 05:21:12.946: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from EXSTART to DOWN, Neighbor Down: Too many retransmissions
R2(config-if)#
*Aug 23 05:22:12.946: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from DOWN to DOWN, Neighbor Down: Ignore timer expired
```

- Messages keep repeating.
- OSPF is clearly not functioning properly.

Fixing the MTU Issue:

```
R2(config-if)#no ip mtu
R2(config-if)#
*Aug 23 05:25:49.362: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
```

Result: Neighbour Reaches FULL State

%OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1

from LOADING to FULL, Loading Done

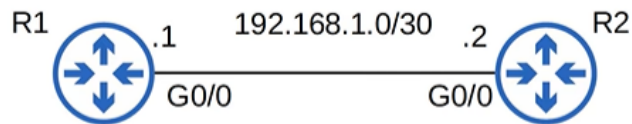
Conclusion (MTU):

- If neighbours cannot reach FULL state:
 - Check MTU settings.

Requirement 8: OSPF Network Type Must Match

Now we look at what happens if OSPF network types don't match.

Configuration Change on R2:



- A loopback interface was configured on R2.
- The loopback was advertised to R1.
- R2's G0/0 network type was changed to point-to-point.
- R1's G0/0 remained broadcast (default).

```
R2(config)#interface lo
R2(config-if)#
*Aug 23 05:52:53.898: %LINK-3-UPDOWN: Interface Loopback0, changed state to up
*Aug 23 05:52:54.898: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#router ospf 1
R2(config-router)#network 2.2.2.2 0.0.0.0 area 0
```

CLI Representation:

```
R2(config-router)#interface g0/0
R2(config-if)#ip ospf network point-to-point
```

R1 G0/0 still using Broadcast Network type

Result:

- %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 from FULL to DOWN, Neighbour Down
- %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 from LOADING to FULL, Loading Done

```
*Aug 23 05:53:34.818: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface
*Aug 23 05:53:34.914: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
R2(config-if)#do show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.1	0	FULL/	00:00:36	192.168.1.1	GigabitEthernet0/0

Neighbour Appears FULL

CLI Representation (R1):

```
R1#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.1.2      1    FULL/BDR        00:00:31   192.168.1.2   GigabitEthernet0/0
```

- Neighbour state is **FULL**.
- It appears OSPF is working correctly.

The Actual Problem:

CLI Representation (R1 Routing Table)

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R1#
```

- R2's loopback address is missing.
- It should be in the routing table.
- OSPF is not operating properly, despite FULL state.

Conclusion (Network Type)

- If OSPF network types don't match:
 - Neighbour state can be FULL
 - Routing information may be missing
- This issue is tricky to troubleshoot
- Always verify network type consistency

Summary: OSPF Neighbour Requirements

- 1) Area number must match
- 2) Interfaces must be in the same subnet
- 3) OSPF process must not be **shutdown**
- 4) OSPF Router IDs must be unique
- 5) Hello and Dead timers must match
- 6) Authentication settings must match

- 7) IP MTU settings must match
- 8) OSPF Network Type must match.

Can become OSPF neighbors, but
OSPF doesn't operate properly.

OSPF LSA Types

Introduction to OSPF LSA Types

The final topic is LSA types.

- LSAs make up the OSPF LSDB.
- All routers in the same OSPF area share the same LSDB.
- There are 11 LSA types.
- For the CCNA, you only need to know three.

Type 1 LSA – Router LSA

- Generated by every router running OSPF.
- Identifies the router using its router ID.
- Lists networks attached to the routers:
 - OSPF-activated interfaces

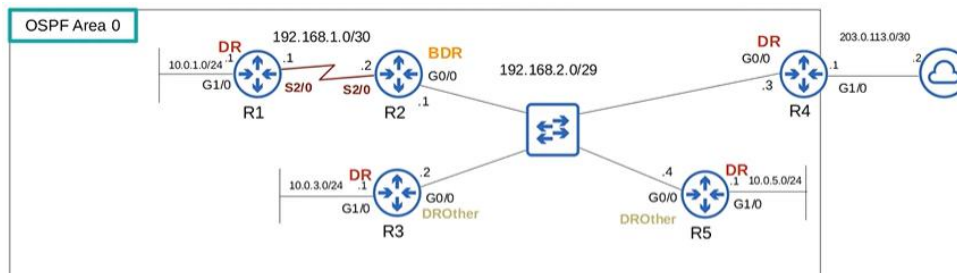
Type 2 LSA – Network LSA

- Generated by the DR.
- Created for each multi-access network.
- Example:
 - Ethernet using the broadcast network type
- Lists:
 - Routers attached to the multi-access network

Type 5 LSA – AS-External LSA

- Generated by ASBRs.
- Describes routes to destinations:
 - Outside the OSPF autonomous system

Viewing the LSDB



CLI Representation:

```

R1#show ip ospf database

        OSPF Router with ID (1.1.1.1) (Process ID 1)

        Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum Link count
1.1.1.1      1.1.1.1     1396        0x80000002  0x00FE8D  4
2.2.2.2      2.2.2.2     932         0x80000005  0x00753F  4
3.3.3.3      3.3.3.3     974         0x80000004  0x00AD70  2
4.4.4.4      4.4.4.4     975         0x80000005  0x004CC2  2
5.5.5.5      5.5.5.5     976         0x80000004  0x00D212  3


        Net Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum
192.168.2.3  4.4.4.4     932         0x80000002  0x00740D


        Type-5 AS External Link States

Link ID      ADV Router   Age         Seq#         Checksum Tag
0.0.0.0      4.4.4.4     273         0x80000002  0x00C0E0  1

```

Observations:

- Each router generated a Type 1 Router LSA.
- Only one Type 2 Network LSA exists:
 - Generated by R4
 - For 192.168.2.0/29
- No Type 2 LSAs for G1/0 interfaces:
 - No other routers connected
- A Type 5 AS-External LSA is generated by R4:
 - Advertising its default route to the Internet

Final Review

- OSPF Network Types:
 - Broadcast
 - Point-to-Point
- OSPF Neighbour & Adjacency Requirements
- OSPF LSA Types:
 - Type 1 (Router LSA)
 - Type 2 (Network LSA)
 - Type 5 (AS-External LSA)